


Designed by Remus Teodorescu 2026

LECTURE 10.2

# Solid Oxide Cells

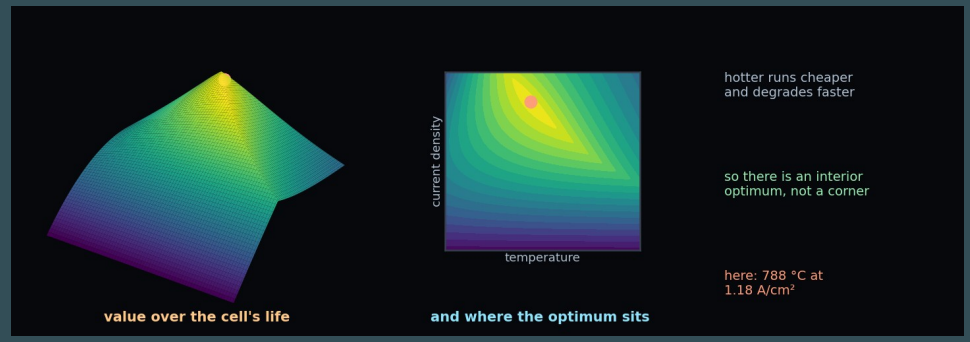

**Remus Teodorescu**

*With support from Research Assistant Noman Khan - nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk


Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME




*Schematic: the value a cell produces over its lifetime against temperature and current, with its best point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the fuel channel, and finds the best voltage by a gradient through one PINN trained over the whole range of voltages.*

# Learning Objectives

## SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam running short, a current that follows the voltage

3. **The best voltage** — one PINN for every voltage, the worth and its gradient

*Incropera carries the transport, Liu the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.*

## READ, AND WHAT EACH ANSWERS

| | |
|---|---|
| **Incropera ch. 1** | where do the losses sit? |
| **Incropera ch. 14** | how does the steam run short? |
| **Liu ch. 5** | what does a gradient buy? |

# Three Electrolysers

**Water electrolyser** — a cell that splits water into hydrogen and oxygen with electricity. The three industrial types differ in their electrolyte, and so in the ion that carries the charge and in the temperature at which they run.

## ONE ELECTROLYTE EACH

**alkaline** — a potassium hydroxide solution near 80 degrees; $OH^-$ carries the charge and a diaphragm keeps the gases apart

**PEM** — a humid polymer membrane near 70 degrees; $H^+$ crosses to the hydrogen side and liquid water is split at the oxygen side

**solid oxide** — a ceramic near 800 degrees; $O^{2-}$ crosses to the air side, steam is split and heat supplies part of the energy

## COST AND CONTROLLABILITY

| | alkaline | PEM | solid oxide |
|---|---|---|---|
| temperature [°C] | 70 – 90 | 50 – 80 | 700 – 850 |
| energy [kWh/kg $H_2$] | 50 – 78 | 50 – 83 | 45 – 55 |
| cost [USD/kW] | 500 – 1000 | 700 – 1400 | stack >2000 |
| minimum load [%] | 10 – 30 | 10 – 40 | about 50 |
| ramp [% per s] | 0.5 – 5 | 3 – 10 | about 0.2 |
| start from cold | minutes | minutes | hours: kept hot |

*Typical values, rounded. The costs are IRENA's 2020 system prices; installed plants cost more: 2000 – 2600 USD/kW outside China in 2024, 600 – 1200 USD/kW in China (IEA, 2025).*



*IRENA (2020); Dorn et al. (2026); IEA (2025)*

# Electrolysis at Industrial Scale

**Power-to-X** — the conversion of renewable electricity into hydrogen by electrolysis, and of the hydrogen into a fuel or chemical, such as methanol, that can be stored and shipped.

## KASSØ, DENMARK, SINCE 2025

| | |
|---|---|
| **electrolysers** | 52 MW of PEM, from Siemens Energy |
| **power** | a 304 MW solar park beside the plant |
| **product** | up to 42 000 t of e-methanol a year |

## INSTALLED WORLDWIDE, AND 2030

| | |
|---|---|
| **installed** | 0.6 GW in 2021, 2 GW in 2024, 4.2 GW in 2025 |
| **by technology** | in 2022: alkaline 60 %, PEM 30 %, SOEC under 1 % |
| **by 2030** | 26 GW committed; 520 GW announced, much now out of reach |
| **China** | 65 % of the capacity installed or committed |




Laura Mærsk, e-methanol fuel
Jens Dohrn, CC BY-SA 4.0




2 MW PEM module, ITM Power
Bubble60, CC BY-SA 4.0

## WHY ALKALINE STILL LEADS

| | |
|---|---|
| **mature** | a century in industry |
| **cheap** | no noble metals, large stacks |
| **China's makers** | alkaline at 600 – 1200 USD/kW |

*Elsewhere 2000 – 2600 USD/kW (IEA). More: Kassø, ITM Power.*



*IEA (2023 – 2026); European Energy*

# The Solid Oxide Electrolyser

**Solid oxide electrolysis cell** — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only $O^{2-}$ ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

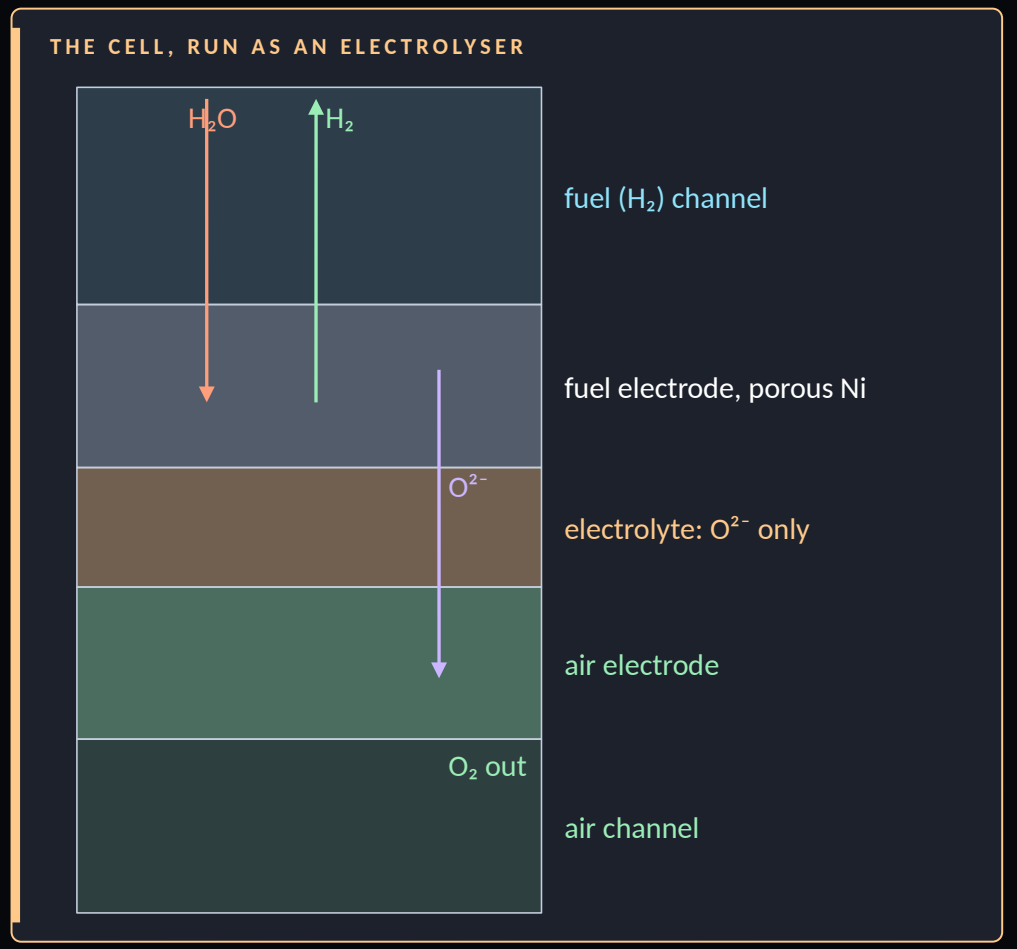
## WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

$\Delta H$ the energy to split water, $\Delta G$ the part that must be electricity, $T\Delta S$ the part heat can supply

## THE THREE LAYERS

| | |
|---|---|
| **fuel electrode** | steam takes two electrons, hydrogen leaves |
| **electrolyte** | carries $O^{2-}$, keeps the gases apart |
| **air electrode** | $O^{2-}$ gives up its electrons as oxygen |



# The Polarisation Curve

**Nernst potential** — the cell voltage at which no current flows, the reversible voltage of the reaction at the local gas composition; it rises as the steam is used up. The polarisation curve is this potential plus the losses, which grow with the current density.

## THE NERNST POTENTIAL, AND THE LOSSES ON TOP

$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

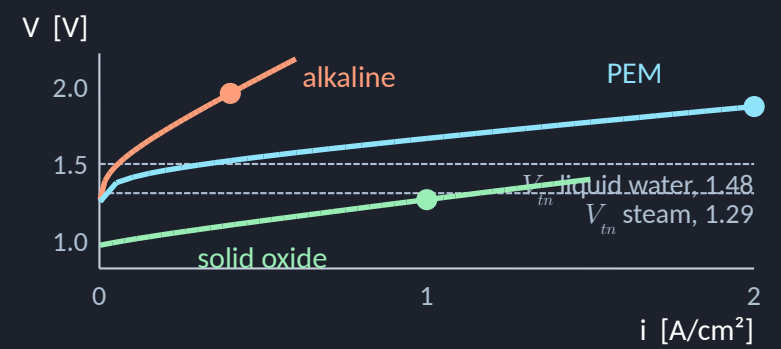
$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

$E$ the Nernst potential, $y$ the steam fraction, $p_{O_2}$ the air side's oxygen, $V$ the cell voltage $\eta_{act}, \eta_{ohm}, \eta_{conc}$ the activation, ohmic and gas-transport losses; in electrolysis they add

The exercise gathers all three losses into one area-specific resistance, 0.25 Ω cm², an estimate that stands for the curve at its operating point.

## THREE CELLS, ENERGY PER KILOGRAM

SCHEMATIC, TYPICAL VALUES




| | |
|---|---|
| **alkaline** | 1.94 V at 0.4 A/cm²: 52 kWh/kg |
| **PEM** | 1.85 V at 2.0 A/cm²: 49 kWh/kg |
| **solid oxide** | 1.25 V at 1.0 A/cm²: 33 kWh/kg |

*26.6 kWh/kg per volt; solid oxide also needs steam heat.*

# The Thermoneutral Voltage

**Thermoneutral voltage** — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero and the temperature stays even. For steam near 800 degrees it is 1.29 volts, and stacks are designed at or just above it.

## THE BALANCE, AND THE HEAT IN ONE TERM

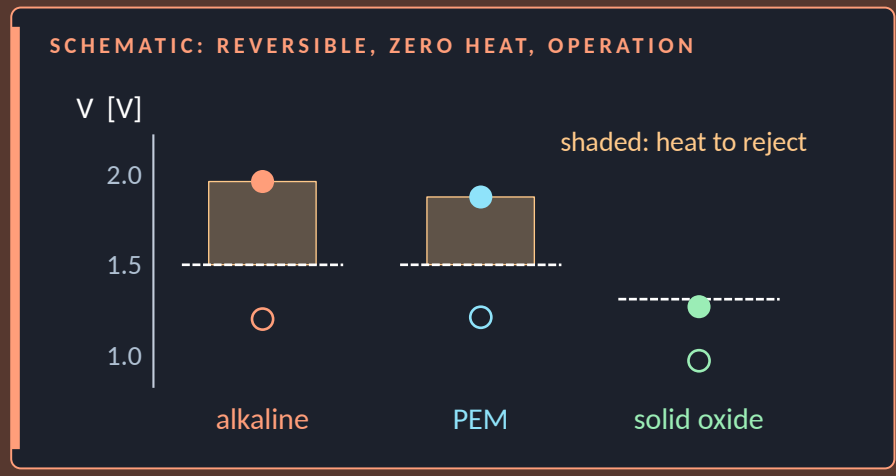
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

$V_{tn}$ the thermoneutral voltage, $\Delta H$ the enthalpy of splitting, $q$ the heat per electrode area, $i$ the current density

**TALK · 2 MIN** How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, why do the two cold cells reject heat while the solid oxide cell runs at its dashed line?

## THREE CELLS ON ONE VOLTAGE AXIS




**alkaline, PEM** 0.4 – 0.5 V above: heat to remove

**solid oxide** at $V_{tn}$: no net heat in the stack

*Liquid water: 1.48 V includes evaporation. Steam at 800 °C: 1.29 V, the steam raised outside.*

# The Fuel Channel

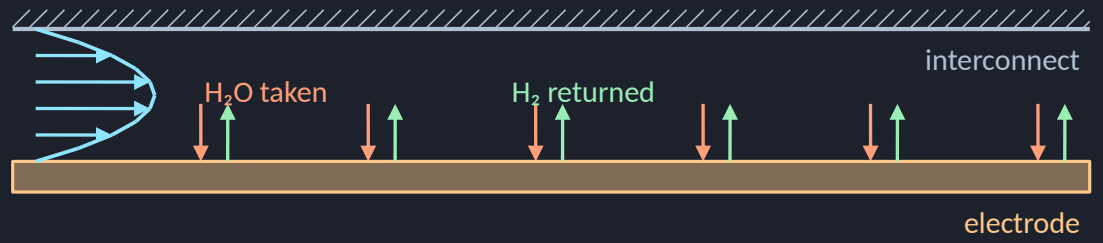
**Fuel channel** — the hydrogen channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s and the electrode takes steam and returns hydrogen, so the steam mole fraction $y$, the share of steam molecules in the gas, falls along the channel.

## THE SCALED EQUATION OF EX_10.2

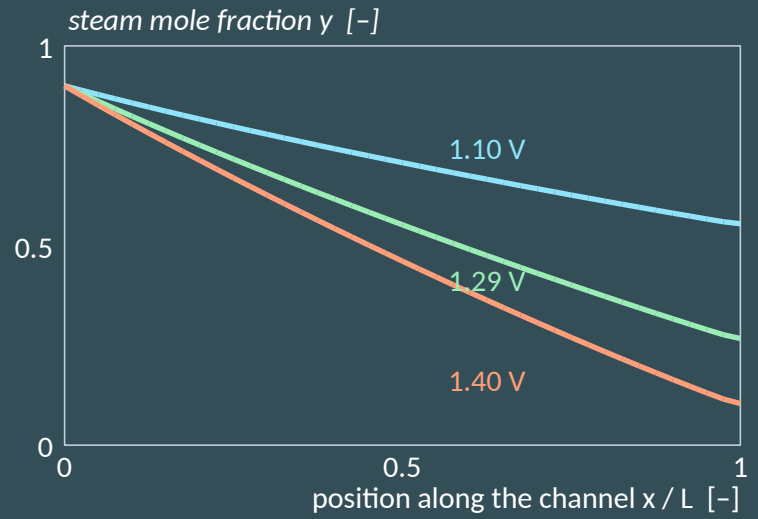
$$y_x = \frac{y_{xx}}{Pe} - K(V - E(y)), \qquad Pe = 125, \quad K = 1.83\ \mathrm{V}^{-1}$$

*y* the steam mole fraction, *x* along the channel over *L*, *Pe* the Péclet number, *K* gathers *L*, *u*, *h*, *c*, *F*, *ASR*

### STEAM TAKEN, HYDROGEN RETURNED




### STEAM ALONG THE CHANNEL




### THE CHANNEL IN THE THREE CELLS

| | fed | the channel must |
|---|---|---|
| alkaline | KOH solution, in excess | carry gas bubbles out |
| PEM | liquid water, in excess | carry bubbles, cool |
| solid oxide | steam, 70 % of it used | keep steam to the outlet |

# The Local Current

**Local current density** — the current per electrode area at one point of the channel. The whole cell is held at one voltage $V$; the local gas sets the local Nernst potential $E(y)$, and their difference over the area-specific resistance is the local current density.

## ONE VOLTAGE, A CURRENT THAT VARIES

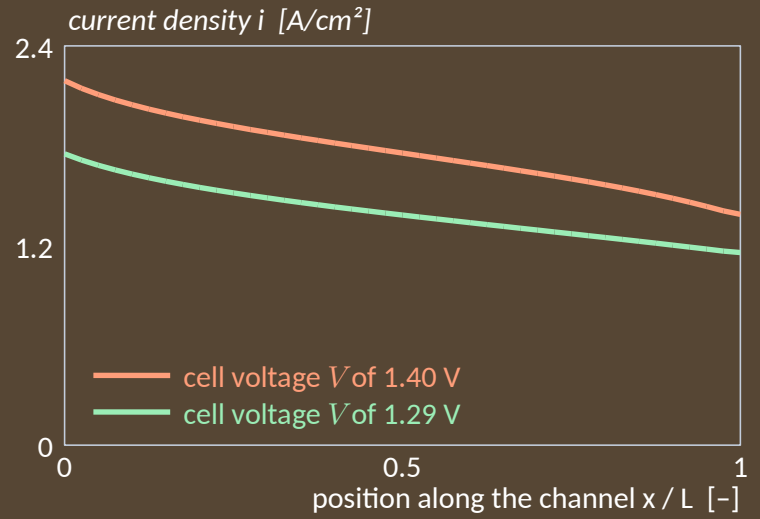
$$i(y) = \frac{V - E(y)}{ASR}$$

$i$ the local current density, $E(y)$ the Nernst potential of the local gas, $ASR$ all losses as one area resistance (an estimate)

## WHY IT FALLS, AND WHAT IT MEANS

| | |
|---|---|
| **1 · steam used** | the steam fraction $y$ falls along the channel |
| **2 · $E$ rises** | the Nernst potential grows as $y$ falls |
| **3 · gap shrinks** | $V$ - $E$, and so $i$, fall toward the outlet |
| **→ hydrogen** | the outlet half makes the smaller share |
| **→ heat** | uneven current, uneven heat, thermal stress |

## THE CURRENT ALONG THE CHANNEL




## THE CURRENT IN THE THREE CELLS

| | along the channel | set by |
|---|---|---|
| alkaline | nearly even | gas bubbles |
| PEM | nearly even | bubbles, membrane |
| solid oxide | 1.75 to 1.16 A/cm² | steam used up |

# Finite Volumes Along the Channel

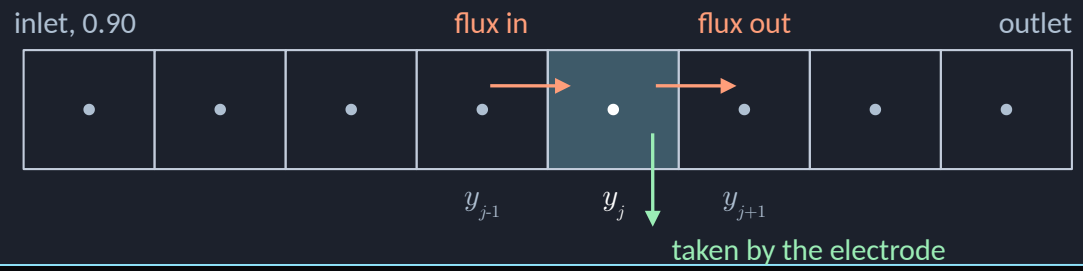
**Finite volumes (FVM)** — the channel cut into cells and the steam balance written for each cell: what the flow and diffusion carry in through one face, less what leaves through the other, equals what the electrode takes. Neighbouring cells share their faces, so the steam is conserved exactly.

## THE BALANCE OF CELL j

$$\frac{y_j - y_{j-1}}{h} - \frac{y_{j+1} - 2y_j + y_{j-1}}{Pe\,h^2} + K(V - E(y_j)) = 0, \qquad j = 1, \ldots, n$$

$y_j$ the steam mole fraction in cell $j$, $h = 1/n$ the cell width: flow from the upstream cell, diffusion between neighbours inlet $y_0 = 0.90$ half a cell before the first centre, outlet zero gradient; $n$ equations, solved by Newton's method

### EIGHT OF THE n CELLS, CELL j PICKED OUT




## REFINED, AGAINST THE REFERENCE

| cells | worst error | |
|---|---|---|
| 25 | 0.0106 | |
| 50 | 0.0042 | meets 0.005 |
| 100 | 0.0019 | |
| 200 | 0.0013 | |

*The error falls about in proportion to the cell width, because the flow term takes the upstream cell. One solve is a few Newton steps on a tridiagonal system, about 2 ms, and it must be repeated for every new voltage.*

# One PINN for Every Voltage

**Parametric PINN** — a PINN trained over two domains at once: the space domain, the position $x$ from inlet to outlet, and the voltage domain, the cell voltage $V$ from 1.10 to 1.40 V. After this one training it gives the steam profile $y(x, V)$ for any voltage in that range.

## THE TRIAL FUNCTION, AND THE LOSS

$$\hat{y}(x, V) = y_{in} + x\,\mathcal{N}(x, V)$$

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{k=1}^{N} r_k^2 + \lambda\,\overline{\hat{y}_x(1, V)^2}, \qquad r = \hat{y}_x - \frac{\hat{y}_{xx}}{Pe} + K(V - E(\hat{y}))$$

$\theta$ the network's weights, $r$ the residual at $N$ collocation points $(x, V)$, $\lambda$ the weight of the zero outlet slope $y_{in} = 0.90$ the inlet, built in, $\mathcal{N}$ the network : $x$ and the cell voltage $V$ in, trained over 1.10 to 1.40 V

### THE TRIAL FUNCTION IN PYTORCH

```
y = Y_IN + x * net(p)              # trial function, p = (x, V)
g = grad(y, p)                     # derivatives by autograd
loss = (r**2).mean() + LAM*out**2  # residual, outlet slope at 0
```

## FVM AGAINST PINN AT 1.29 V

| | FVM, 50 cells | PINN |
|---|---|---|
| worst error | 0.0042 | 0.0005 |
| one solution | 2 ms | under 1 ms |
| a new voltage | solve again | read off |
| $dC/dV$ | by finite steps | by autograd |
| training, offline | none | 20 s |

## WHY A PINN HERE

**the task** the steam for every voltage

**FVM** one solve per voltage

**PINN** any voltage, and its gradient

*FVM as on the previous slide, with 50 cells.*

# Optimisation of the Cell Voltage

**Cost function** — the number the operator minimises: the cost of the electricity bought less the value of the hydrogen sold, per hour, as a function of the cell voltage $V$. The prices are assumptions, 3 euro per kg and 0.065 euro per kWh, so the optimum is conditional on them.

## THE COST, AND ITS MINIMUM THROUGH THE PINN

$$C(V) = p_{el}\,(V - V_h)\,I(V), \qquad V_h = \frac{p_{H_2} M_{H_2}}{2F\,p_{el}} = 1.74\ \text{V}$$

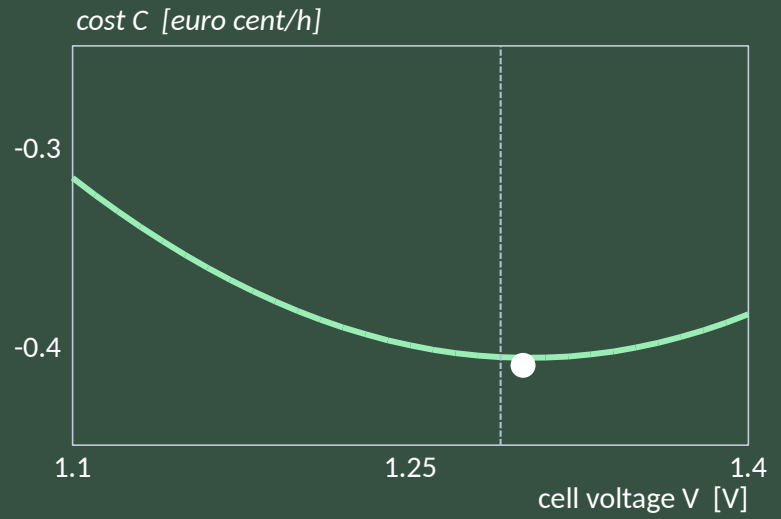
$$I(V) = A \int_0^1 \frac{V - E(\hat{y}(x, V))}{ASR}\,dx, \qquad V^* = \arg\min_V\ C(V)$$

$C$ the cost of an hour (negative : a profit), $p_{el}, p_{H_2}$ the prices (assumptions), $V_h$ where hydrogen pays the electricity
$I$ the total current, $A$ the electrode area, $\hat{y}$ the PINN's steam fraction, $V^*$ the best voltage

### THE BEST VOLTAGE IN PYTORCH

```
V = torch.tensor([1.25], requires_grad=True)  # the decision variable
C = cost(net, V)                 # steam from the PINN, current, then the cost
V = V - lr * grad(C, V)[0]       # downhill, through the network: 200 steps
```

## THE COST AGAINST THE VOLTAGE




| | |
|---|---|
| **the PINN gives** | the steam profile $y(x,\ V)$ |
| **the optimiser** | $V^*$ = 1.30 V, the reference |
| **found in** | 0.2 s; by 61 solver runs, 4 s |

# No Best Point Over a Lifetime

**Degradation** — the slow rise of the cell's area-specific resistance over thousands of hours, from physical changes in its layers (right). The rate law below is an empirical approximation fitted to long tests, not derived from those mechanisms; published rates span an order of magnitude.

## THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

$$\frac{d\,ASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\min_{V(t),\,T(t)} \int_0^{t_{life}} [p_{el} P - p_{H_2} \dot{m}_{H_2}]\,dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

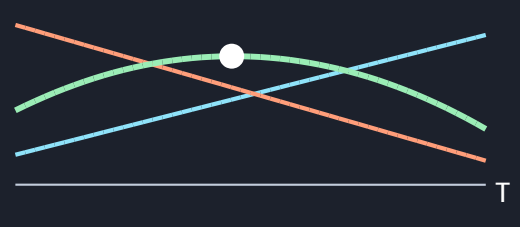
$k_0, E_d, n$ fitted to long tests : an empirical law, Arrhenius in $T$ and a power of $i$, not derived from the mechanisms

**TALK · 2 MIN** How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

## WHAT AGES, AND WHY

| layer | mechanism |
|---|---|
| fuel electrode | nickel grains coarsen and migrate: fewer reaction sites |
| air electrode | oxygen pressure at the interface lifts it off |
| steel, air side | chromium vapour deposits and blocks sites |

THE TRADE, SCHEMATIC




cent/h: hotter is cheaper

hours left: fewer

product: a best inside

**so the answer** a trajectory $V(t), T(t)$

# Operating for Cost and Lifetime

**Operating strategy** — the cell voltage $V(t)$ and temperature $T(t)$ chosen over the life of the stack to minimise the cost of each kilogram of hydrogen: the electricity bought hour by hour, and the stack replaced when its resistance reaches the end-of-life limit.

## THE LEVELISED COST OF HYDROGEN

$$\mathrm{LCOH} = \frac{C_{stack} + \sum_t p_{el}(t)\, P(t)\, \Delta t}{\sum_t \dot{m}_{H_2}(t)\, \Delta t} \quad \text{over the stack's life}$$

LCOH the cost of a kilogram of hydrogen, $C_{stack}$ the stack, replaced at end of life, $p_{el}(t)$ the hourly price $P(t)$, $\dot{m}_{H_2}(t)$ power and hydrogen at the chosen $V(t), T(t)$; the life ends when $ASR$ reaches $ASR_{\max}$

## THE TRADE-OFFS

| | |
|---|---|
| **higher voltage** | more hydrogen per hour, faster ageing |
| **higher temperature** | lower resistance, faster ageing |
| **cheap hours** | run harder when the price is low |
| **near $V_{tn}$** | small heat, small gradients in the ceramic |

## WHAT THE PINN ADDS

| | |
|---|---|
| **one hour** | the cost in a millisecond |
| **gradient** | $dC/dV$ and $dC/dT$ by autograd |
| **a solver** | one solve per variable per step |
| **a lifetime** | an optimiser over $V(t), T(t)$ |
| **Ex_10.2** | 0.2 s against 4 s of sweeps |

```
C = lcoh(net, V_t, T_t)           # cost over a life
V_t = V_t - lr * grad(C, V_t)[0]  # schedule, downhill
```

*Degradation rates are uncertain tenfold: a predicted lifetime ranks strategies, it is not a promise.*

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic | what it adds | read | where it makes a difference |
|---|---|---|---|
| channel temperature | the cell cools below $V_{tn}$ | Incropera ch. 5 | stacks; MP10.2A |
| degradation law | $k_0$, $E_d$, $n$ fitted to drifts | Liu ch. 5 | lifetime; MP10.2B |
| button-cell fit | kinetics from measured curves | Incropera ch. 1 | validation |
| daily schedule | the hourly price as driver | Liu ch. 5 | grid operation; L12 |
| steam supply | outlet gas heat raises the steam | Incropera ch. 11 | plant efficiency |

### WHY THEY ARE LEFT OUT

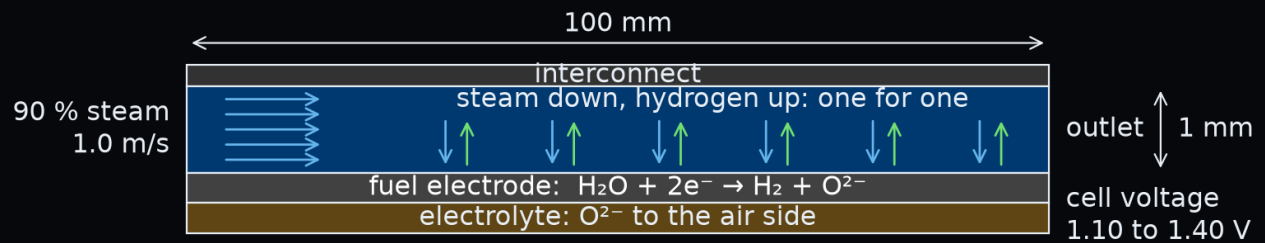
Ex_10.2 solves one isothermal channel at one moment, scored against a boundary-value solver, so none of them is needed to do it. The first two rows are the two mini projects; the last two are where the schedule meets the grid of L12.

EXERCISE

# Exercise 10.2

## ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its channel, the Nernst potential and $K$
- **4** — the boundary-value reference; finite volumes with Newton
- **5** — one PINN for every voltage from 1.10 to 1.40 V
- **6** — the best cell voltage, by a gradient through the network
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B




## THE QUESTION EACH PART ANSWERS

| | |
|---|---|
| **1 – 3** | what sets the local current? |
| **4** | how fine must the grid be? |
| **5** | one network, every voltage? |
| **6** | which voltage pays best? |

## REPORT QUESTION, TO CONCLUDE

The ASR, the diffusivity and the prices are the notebook's estimates, and two students with different price ratios should find different best voltages. What in your answer is physics, and what is an assumption?

SUMMARY

# Key Takeaways

| SOLID OXIDE CELL | |
|---|---|
| **what it is** | a ceramic cell, two directions |
| **function** | steam to hydrogen, heat helping |
| **train, tune** | run at or just above $V_{tn}$ |
| **vs low temperature** | a quarter less electricity |
| **used for** | storing surplus power as fuel |

| LIFETIME OPTIMISATION | |
|---|---|
| **what it is** | a PINN with the voltage as input |
| **function** | the cost, and its gradient |
| **train, tune** | once, over 1.10 to 1.40 V |
| **vs a solver** | the gradient comes free |
| **used for** | a trajectory, not a setting |

*Next — L11.1: a network in a real-time loop, where there is no PDE at all.*

# Questions

## 1 – 2 · A NETWORK FOR THE FUEL CHANNEL

**1** **How can a network learn the steam fraction along the channel of the cell for every cell voltage?**

The network takes the position along the channel and the cell voltage, and is trained over 1.10 to 1.40 V until the species equation holds at points along the channel, with the local current set by the gap between the cell voltage and the local Nernst potential. The steam fraction at the inlet is built into the form of the solution, and the zero slope at the outlet is held by a loss term. After one training, any voltage in the range is read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

**2** **Why does the local current make the fuel channel harder to solve, and how is the network's answer checked?**

The flux at the electrode is no longer given: the current at each point follows the local steam fraction, so the equation is nonlinear in it and needs a condition at the outlet, and it can no longer be marched from the inlet. The network takes the nonlinear term into its residual with no change of method. Its answer is scored against a boundary-value solver at a tight tolerance: at 1.29 V within 0.0005 of the steam fraction, where finite volumes on 50 cells are within 0.0042. *Ex_10.2 sections 3 – 5*

## 3 – 4 · THE BEST VOLTAGE

**3** **How can the best cell voltage be found with a network, and what does the gradient save?**

The worth of an hour, the hydrogen made at its price less the electricity used at its price, is computed from the network's steam fraction and current. With the voltage as an input, automatic differentiation gives the derivative of the worth with respect to it, and gradient ascent finds 1.30 V in 200 steps and 0.2 s; a sweep of 61 runs of the reference solver finds the same voltage in 4 s. The prices are assumptions, so the voltage found is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

**4** **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of the voltage and the temperature, because what produces more now ages the cell faster. A solver gives the worth, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives the gradient with respect to every input by automatic differentiation. Degradation rates are uncertain by an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

NEXT

# L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

## BEFORE THE NEXT SESSION

| | |
|---|---|
| **read** | Liu ch. 5 |
| **run** | Ex_10.2, sections 1 to 6 |
| **bring** | your best voltage, and its worth |
| **think about** | what was physics, what was price |

*Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.*